

# From Grammar Descriptors to Knowledge-Tracing Benchmarks: An Auditable Grammar-KT Pipeline

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## Abstract

Knowledge tracing (KT) assumes an item-knowledge-component (KC) mapping, but proficiency resources commonly state grammatical targets in prose. We present Grammar-KT, an auditable pipeline from grammar descriptors to canonical representations, generated items, Q-matrices, and synthetic chronological KT data. In a clean run over 139 English Grammar Profile descriptors, 44 yielded complete mappings to 24 unique six-dimensional grammar cells, all eight repeated annotation units produced identical outputs. Factorized, interaction-augmented, and whole-cell policies produced respectively 9, 14, and 24 KCs, with sharply different Q-matrix structures. Deterministic checks and an independent automated diagnostic accepted 45/45, 69/70, and 120/120 generated items. Across five simulation seeds, a logistic baseline had the best test AUC and log loss within every policy-specific dataset. Because the simulator uses the evaluated KC assignments and each policy induces a different item pool, these scores are technical integration checks rather than evidence that one representation better models human learning. The main contribution is therefore a provenance-preserving benchmark construction method that exposes, rather than hides, consequential representation choices.

## 1 Introduction

Knowledge tracing (KT) estimates a learner's changing knowledge from a chronological sequence of item responses (Corbett and Anderson, 1995; Abdelrahman et al., 2023). Its inputs normally associate each item with one or more knowledge components (KCs), often through a binary Q-matrix. That matrix presupposes decisions that are easy to hide: what counts as a skill, how a curriculum description becomes that representation, and whether an item really exercises the selected components. These are scientific choices, not file formatting.

Grammar resources pose a sharp version of the problem. The English Grammar Profile (EGP) contains over 1,200 corpus-derived competence descriptions mapped to CEFR levels (O’Keeffe and Mark, 2017). A description can combine tense, aspect, voice, polarity, clause type, and modality, state bounded alternatives, or fall outside a selected ontology. Treating every descriptor as one KC obscures reusable operations; decomposing it without explicit rules can invent unsupported distinctions or erase the source connection.

We introduce Grammar-KT, a research harness that makes this path explicit. Figure 1 separates source selection, model-assisted normalization, canonicalization, realization, KC projection, item construction, Q-matrix derivation, simulation, and KT evaluation. Prompts, rulebooks, policies, templates, and parameters remain replaceable versioned artifacts.

We ask:

- RQ1 How completely and consistently can EGP prose be normalized into a closed grammar representation?
- RQ2 How do factorized, interaction-augmented, and whole-cell policies change item-KC structure?
- RQ3 Do all three resulting datasets support reproducible end-to-end KT checks, and what can those checks legitimately show?

Our contributions are a conservative six-dimensional representation, a provenance path from prose to Q-matrix entries, and an executable comparison of three KC policies. Unlike a human-data evaluation, this study cannot establish that any KC is cognitively atomic or that predictive differences transfer to learners; technical integrity and educational validity are kept separate.

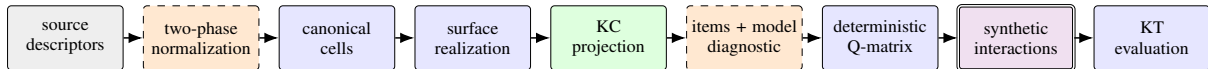


Figure 1: The Grammar-KT pipeline. Model assistance is confined to normalization and an independent item diagnostic; canonicalization, realization, KC rule evaluation, item construction, and Q-matrix derivation have deterministic executors. Simulation is explicitly separated from claims about human learner data.

## 2 Related Work

**Proficiency-scale grammar resources.** The CEFR provides proficiency scales across language activities (Council of Europe, 2020). The EGP specializes this programme for English grammar, using learner-corpus frequency, accuracy, and dispersion to associate competence statements with CEFR levels (O’Keeffe and Mark, 2017). Recent work uses NLP to quantitatively re-examine EGP level assignments (Verratti-Souto et al., 2025). We do not reassess those levels. Our problem is complementary: converting descriptor content into a bounded structural representation while preserving uncertainty and provenance.

**Exercise generation for language learning.** Adaptive question generation has conditioned generation on learner history and target difficulty (Srivastava and Goodman, 2021; Cui and Sachan, 2023). Grammar-focused systems generate multiple-choice or gap-filling exercises from passages or examples (Chan et al., 2022; Bitew et al., 2023). These studies highlight both the cost of authoring and the need to evaluate correctness and answer uniqueness. Our controlled transformations sacrifice open-ended diversity for an exact, replayable relation among a cell, its realization, and its KC labels.

**Domain models and knowledge tracing.** KCs describe acquired units inferred from performance on related tasks (Koedinger et al., 2012). Classical BKT tracks a latent learned/unlearned state with learning, guess, and slip parameters (Corbett and Anderson, 1995); factor models instead use logistic predictors and practice features (Pavlik et al., 2009). A Q-matrix supplies the assumed item-KC mapping, but expert matrices can be refined or learned from response data (Pardos and Dadu, 2018). Picones et al. (2022) combine domain-model induction and student modelling in one pipeline. In contrast, we derive the domain model from linguistic specifications before responses exist, then freeze it before simulation. Synthetic KT data have supported controlled model tests (Piech et al., 2015;

Pagonis et al., 2024); here they test software behavior, not the cognitive truth of the generating policy.

## 3 Task Formulation and Source Data

### 3.1 Grammar-cell representation

For descriptor  $d$ , normalization returns a status, cells, and a note. A cell has six dimensions,

$$c = (t, a, v, p, q, m), \quad (1)$$

for tense, aspect, voice, polarity, clause type, and modal choice. Table 1 gives the domains. Scalars record exact evidence, lists bounded unresolved alternatives, and null no usable constraint. Dimensions within a cell are conjunctive; separate cells are source-supported alternatives. Lists are not expanded into unsupported Cartesian products.

Mappings are *complete* only when all values are scalar. Partial mappings contain a list or null. Zero-cell statuses distinguish out-of-scope, schema-failure, and unresolved cases. Compatibility constraints include  $m \neq \text{none} \Rightarrow t = \text{NA}$  and imperative clauses requiring  $t = \text{NA}$  and  $m = \text{none}$ . Only complete cells enter the downstream canonical inventory.

### 3.2 Declared source subset

The input is a consult-only parsed EGP snapshot that is not redistributed. Its declared SHA-256 digest and size are checked before execution. From 1,222 records, an ordered list selects 139 descriptors: 23 regression controls and 116 fresh cases stratified over tense-aspect combinations, passive voice, negation, questions, imperatives, central modals, and out-of-scope phenomena.

The design contains 139 primary annotation units and eight blinded repeats. Every unit retains model evidence; only primaries feed canonicalization.

Phase 1 exposes the identifier, category fields, guideword, and can-do statement. Source examples remain hidden unless a partial result is eligible for Phase 2, preventing examples from silently replacing the descriptor as the definition of coverage.

| Dimension | Symbol | Closed scalar domain                                                | Principal constraint or interpretation             |
|-----------|--------|---------------------------------------------------------------------|----------------------------------------------------|
| Tense     | $t$    | present, past, NA                                                   | NA is structural and differs from missing evidence |
| Aspect    | $a$    | none, progressive, perfect, perfect–progressive                     | Encodes auxiliary-chain aspect                     |
| Voice     | $v$    | active, passive                                                     | Passive is canonical BE-passive only               |
| Polarity  | $p$    | positive, negative                                                  | Verbal polarity, not lexical meaning               |
| Clause    | $q$    | declarative, polar question, subject-WH, non-subject-WH, imperative | Question subtypes and imperatives are distinct     |
| Modal     | $m$    | none; can, could, may, might, must, shall, should, will, would      | A central modal requires $t = \text{NA}$           |

Table 1: The six dimensions of a scalar GRAMMARCELL. During partial normalization, a dimension may instead hold a finite list or null; only scalar cells enter canonicalization.

## 4 Grammar-to-KT Pipeline

### 4.1 Normalization and canonicalization

Each annotation unit is processed in a fresh model context containing the closed schema, rulebook, phase instructions, and JSON schema. A validator checks fields, domains, status invariants, and tense–modal and imperative compatibility. Every attempt retains its prompt, raw and parsed output, invocation metadata, and errors.

Only a partial Phase 1 result can normally enter Phase 2. Its note names the non-scalar dimensions eligible for refinement; Phase 2 additionally sees the source examples. The transition validator freezes exact and ineligible dimensions, requires every new cell to descend from a Phase 1 cell, and forbids unlicensed list expansion. Contradictory evidence produces an unresolved result rather than silently changing an exact value.

Complete mappings are serialized in a fixed dimension order and assigned a content-derived identifier. Equal tuples therefore deduplicate independently of source identity or order. A canonical inventory stores each tuple once; an edge table separately records every source mapping cell, its descriptor and position. Deduplication consequently preserves many-to-one provenance.

### 4.2 Realization and KC policies

The realizer pairs each canonical cell with a validated predicate frame, subject features, and any WH or imperative conditions. A ten-frame lexicon supplies inflectional forms and argument structure. An ordered procedure constructs modal, perfect, progressive, passive, and lexical chains before applying negation, DO-support, inversion, WH-fronting, or imperative rules. The retained

derivation includes tokens, agreement site, auxiliary chain, and named operations.

A KC opportunity joins a canonical cell, one realization, its operations, split, and source provenance. We compare three declarative policies. The *factorized* policy names marked reusable decisions such as finite tense, aspect, passive voice, negation, questions, imperatives, and individual modals. The *interaction* policy adds five named combinations: DO-support in questions and negation, perfect–progressive, progressive–passive, and perfect–passive chains. The *whole-cell* policy assigns one KC to each exact six-dimensional cell. It is a high-resolution structural baseline, not a cognitive claim. Every policy materializes a frozen cell–KC projection; leaving a cell without a KC is fatal.

### 4.3 Items and Q-matrices

For every activated KC, the constructor searches eligible cells, source conditions, predicate frames, and subjects in stable order to create five unique controlled-transformation items. A prompt exposes the intended grammar and fixed lexical material; the accepted answer is the deterministic realization. The first two replicates are development items and the remaining three are held out.

Deterministic checks cover record shape, provenance, realization, punctuation, uniqueness, and agreement with the frozen projection. Passing candidates receive an independent automated diagnostic for plausibility, naturalness, unsupported constructions, world-knowledge dependence, and answer ambiguity. This diagnostic is not human validation. For accepted item  $i$  and KC  $k$ ,

$$Q_{ik} = \mathbb{I}[k \in \mathcal{K}(c_i)], \quad (2)$$

where  $\mathcal{K}(c_i)$  is the frozen projection of the item’s cell. Unknown identifiers, empty rows, and label-projection disagreements are fatal; density, row width, repeated columns, and low support remain reportable diagnostics rather than triggers for hidden repair.

For example, the source descriptor for affirmative lexical yes/no questions maps to a cell with present tense, no aspect, active voice, positive polarity, a polar-question clause, and no modal. The canonicalizer assigns that tuple a content-derived identifier; one generated item is “Does the technician review the proposal?”; and the factorized projection activates finite-present and polar-question KCs. The edge record retains the original descriptor throughout this chain.

## 5 Experimental Design

The base run uses the factorized policy. Two variants reuse its frozen source, normalization, canonical, and realization artifacts, changing only the KC policy and downstream stages. Because construction makes five items per KC, the resulting item pools are policy-specific; Q-matrix structure is directly comparable, whereas predictive scores are not a controlled estimate of a policy effect.

The simulator is a transparent integration test. It draws independent initial mastery for 60 learners from each of three profiles with Beta(2, 6), Beta(2, 2), and Beta(6, 2) priors. Every learner receives two shuffled passes over the relevant accepted item pool. For active KCs  $A_i$ , mastery  $m_{lk}$ , deterministic item difficulty  $d_i \in [-0.8, 0.8]$ , and multi-KC penalty 0.15, the response logit is

$$z_{li} = |A_i|^{-1} \sum_{k \in A_i} \text{logit}(m_{lk}) - d_i - 0.15(|A_i| - 1), \quad (3)$$

and  $p_{li} = 0.1 + 0.8\sigma(z_{li})$ . Mastery increases after every response at a profile-specific rate. Each learner is split chronologically 60/20/20. Latent mastery, probabilities, random draws, and profile labels are written to separate oracle files and rejected from KT input.

We evaluate a Beta-smoothed empirical KC rate, fixed-parameter BKT ( $P(T) = .04$ ,  $P(G) = .20$ ,  $P(S) = .10$ ), and  $L_2$ -regularized logistic regression. The latter uses only pre-event learner success, active-KC success, prior opportunities, item difficulty, KC count, and one-hot KC indicators. We run seeds 20260817–20260821 and report mean test AUC and log loss with 95%  $t$  intervals over

| Status         | Phase 1 | Final |
|----------------|---------|-------|
| Complete       | 42      | 44    |
| Partial        | 80      | 77    |
| Out of scope   | 16      | 16    |
| Unresolved     | 1       | 2     |
| Schema failure | 0       | 0     |
| Total          | 139     | 139   |

Table 2: Primary normalization outcomes before and after Phase 2. Only the 80 partial Phase 1 cases were eligible for refinement.

the five runs. Accuracy and validation metrics are retained in the artifact but are secondary here.

## 6 Results

### 6.1 Normalization and stability

Phase 1 returned 42 complete, 80 partial, 16 out-of-scope, and one unresolved primary mapping. Of the 80 partial cases routed to Phase 2, two became complete, 77 remained partial, and one became unresolved. The final outcome was therefore 44/139 complete mappings (31.7%), 77 partial, 16 out-of-scope, two unresolved, and no schema failures. Conservative filtering admitted 48 source-cell edges into 24 unique canonical cells. Thirteen cells had one supporting descriptor, six had two, four had four, and one had seven.

All eight repeated annotation units matched their primary unit exactly in status, cells, and note. The five repeated item diagnostics likewise agreed on all decision flags, although their free-text notes differed. These small repeat sets show deterministic-looking behavior for the sampled calls, not a general reliability estimate.

### 6.2 Policy and item structure

Table 3 shows that policy choice materially changes the benchmark. Factorization shares nine KCs across the 24 cells and yields a mean row width of 2.20. Adding interactions increases this to 14 KCs and 2.91; the DO-question interaction is identical to the polar-question column on this sample, so it adds no observed distinction. Whole-cell encoding produces 24 orthogonal columns and unit-width rows.

Deterministic checks rejected no candidate. The automated diagnostic accepted all 45 factorized and all 120 whole-cell items. It rejected one of 70 interaction items for suspected answer ambiguity: the negative past perfect-progressive target placed *not*

| Policy         | KCs | Candidates | Accepted | $Q$ edges | Density | Row widths      |
|----------------|-----|------------|----------|-----------|---------|-----------------|
| Factorized     | 9   | 45         | 45       | 99        | .244    | 1–4 (mean 2.20) |
| + interactions | 14  | 70         | 69       | 201       | .208    | 1–5 (mean 2.91) |
| Whole cell     | 24  | 120        | 120      | 120       | .042    | 1 (mean 1.00)   |

Table 3: Policy-specific item and Q-matrix outcomes. Density is the fraction of active entries. Item pools differ because five candidates are constructed per activated KC.

after the first auxiliary, while the diagnostic identified a marked alternative placement consistent with the literal prompt. This case illustrates why deterministic realization alone does not establish a unique pedagogically acceptable answer.

### 6.3 Synthetic KT checks

Every run passed Q-matrix integrity and observable/oracle separation checks. The policy-specific simulations contained 16,200, 24,840, and 43,200 events per seed for factorized, interaction, and whole-cell policies. Within each dataset, logistic regression achieved the highest mean test AUC and lowest log loss (Table 4); fixed-parameter BKT had the highest log loss. These results establish that the generated artifacts support chronological KT evaluation and respond consistently across seeds. They do not rank the KC policies: the item counts, row widths, response-generating states, and outcome rates all change with the policy.

## 7 Discussion

**RQ1: conservative normalization exposes the bottleneck.** Only 31.7% of sampled descriptors supported exact cells. Phase 2 resolved two of 80 partial cases, so examples rarely licensed a safe refinement under the transition rules. This is not merely low recall: retaining 77 partial mappings prevents the system from manufacturing combinations that the source does not establish. Exact agreement on eight repeats is encouraging but too small, and too dependent on one unpinned model service, to substitute for expert review.

**RQ2: policies encode different structural assumptions.** Factorized KCs reuse evidence but produce multi-KC rows. Whole cells eliminate overlap at the cost of 24 narrowly supported categories. Interactions increase row width and, in one case, duplicate an existing column on the observed domain. The harness makes such redundancy visible before a representation is used with learners.

**RQ3: integration success is not cognitive validation.** All policy artifacts support leakage-audited

simulation and three KT baselines; within each, logistic regression is the strongest technical baseline. But the simulator defines mastery and responses using the same KC assignments later given to the models. Moreover, five items per KC creates different datasets. Predictive differences are therefore conditional on each synthetic world, not evidence that a policy captures human acquisition. A valid policy comparison requires a shared, independently labelled item set and real learner sequences.

## 8 Reproducibility and Audit Trail

Prompts, rulebooks, schemas, lexicons, policies, templates, and parameters are versioned separately from their executors. Each run stores the resolved experiment, source digest, seed, Git commit and dirty state, explicit reuse, stage summaries, and per-call prompts and responses. The three primary runs used clean snapshot commit d58834e; seed variants used dcfb8e8. Earlier stages were copied only through the declared parent-and-start-stage mechanism. The two-commit clean history is retained as runs/evidence\_snapshot.bundle. Exact external reruns remain limited because the consult-only EGP snapshot is not redistributed and the configured model family does not expose a pinned snapshot or all decoding parameters.

## Limitations

The ontology covers single-clause English verbal morphosyntax: present and past tense, four aspect values, active and canonical BE-passive voice, polarity, five clause types, and nine central modals. It excludes question tags, linked conditionals, embedded questions, GET-passives, semi-modals, non-verbal grammar, and multi-clause relations. The stratified 139-descriptor sample is not the full EGP and only exact mappings enter downstream stages.

Normalization and item diagnostics use one automated model backend without expert adjudication. Eight normalization repeats and five diagnostic repeats are too few for general reliability claims. The diagnostic’s near-universal acceptance



| Policy-specific data | Model     | Test AUC $\uparrow$               | Test log loss $\downarrow$        |
|----------------------|-----------|-----------------------------------|-----------------------------------|
| Factorized           | Empirical | .613 $\pm$ .010                   | .661 $\pm$ .010                   |
|                      | BKT       | .603 $\pm$ .007                   | .723 $\pm$ .016                   |
|                      | Logistic  | <b>.658 <math>\pm</math> .013</b> | <b>.618 <math>\pm</math> .010</b> |
| + interactions       | Empirical | .608 $\pm$ .008                   | .649 $\pm$ .006                   |
|                      | BKT       | .603 $\pm$ .014                   | .690 $\pm$ .007                   |
|                      | Logistic  | <b>.646 <math>\pm</math> .012</b> | <b>.603 <math>\pm</math> .005</b> |
| Whole cell           | Empirical | .659 $\pm$ .005                   | .661 $\pm$ .005                   |
|                      | BKT       | .650 $\pm$ .005                   | .729 $\pm$ .006                   |
|                      | Logistic  | <b>.691 <math>\pm</math> .009</b> | <b>.626 <math>\pm</math> .006</b> |

Table 4: Mean test metrics and 95%  $t$  intervals over five simulation seeds. Bold marks the best model within each policy-specific dataset. Scores must not be compared as controlled policy effects because the item pools and generating states differ.

is consistency evidence, not proof of naturalness or uniqueness; the model snapshot and all decoding parameters are not pinned. Surface realization also uses a ten-frame lexicon and fixed transformations, limiting lexical and discourse diversity.

Most importantly, the simulator defines responses from the same KC assignments being evaluated, and the five-items-per-KC rule gives each policy a different item pool. Its outputs can expose leakage, sparsity, and software failures, but cannot validate a policy’s cognitive reality, acquisition order, or transfer to learners. Such claims require expert-labelled shared items and independent human interaction data.

## Ethics Statement

No learner or human-participant records are used; all sequences and profile labels are synthetic. Profile names denote Beta-distribution settings, not demographic or ability groups. Generated items must not be deployed for assessment or instruction without human review for correctness, ambiguity, accessibility, and bias. Use and quotation of the external EGP source remain subject to its licence; the artifact stores identifiers, selection metadata, and verification material rather than redistributing the snapshot.

## 9 Conclusion

Grammar-KT turns prose grammar descriptors into inspectable KT artifacts while preserving ambiguity and provenance. Fresh runs show that normalization is the main coverage bottleneck and that plausible KC policies yield substantially different Q-matrices. All three support reproducible synthetic KT checks, but those checks cannot select a cognitively valid representation. The next step

is expert validation followed by evaluation on a shared item pool and human learner sequences.

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## A Artifact Map

Table 5 maps the principal conceptual choices to their repository artifacts. The map is intended to keep the paper narrative aligned with executable inputs as the experiment evolves.

| Component       | Scientific artifacts and executor                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| Source          | sample IDs, metadata, annotation units, and digest; source.py                                                                      |
| Normalization   | phase prompts, grammar domains, rulebook, JSON schema, and backend configuration; normalisation.py and normalisation_validation.py |
| Canonical cells | fixed dimension order and record constraints; canonical.py and records.py                                                          |
| Realization     | lexical frames, rules, and held-out cell list; realisation.py                                                                      |
| KC projection   | factorized, interaction, and full-cell policies; kc.py                                                                             |
| Items           | transformation template and diagnostic criteria; items.py and item_validation.py                                                   |
| Q-matrix        | frozen cell–KC projection and integrity rules; qmatrix.py                                                                          |
| Simulation      | learner-profile and response parameters; simulation.py                                                                             |
| KT              | empirical, BKT, and logistic settings; kt.py                                                                                       |

Table 5: Conceptual components, replaceable scientific artifacts, and their executors. Paths are relative to the artifact repository.

## B Evidence Retained for the Reported Runs

The archived run directories retain resolved configurations, clean source commit identifiers and digests, reuse boundaries, stage summaries, prompts, raw and parsed model responses, validation decisions, Q-matrices, simulated event tables, fitted-model predictions, and metric summaries. The primary run and each policy variant use five declared simulation seeds. Human evaluation is not part of the present evidence; adding it would require a pre-declared sample, qualified annotators, instructions, adjudication, agreement measures, and uncertainty reporting.